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Mutual Fund Analytics

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Author: Sk Ali Abbas

Comprehensive project report presenting data-driven analytics, performance measurement, risk assessment, and investor behavior analysis for mutual fund portfolios. This report synthesizes methodology, findings, and actionable recommendations for portfolio managers and stakeholders.

Sk Ali Abbas

Executive Summary

The Mutual Fund Analytics project consolidates multi-source investment data to evaluate fund performance, quantify risk, and profile investor behavior. Emphasis was placed on rigorous data cleaning, robust performance metrics (CAGR, Sharpe, Sortino, Max Drawdown), risk measures (VaR, CVaR, rolling Sharpe), and cohort-based investor analytics including SIP continuity. Findings identify top-performing strategies, risk concentrations, and actionable product improvements. The report concludes with recommendations for product, risk, and analytics teams and a roadmap for operationalizing insights in Power BI.

- Scope: Cross-sectional fund and investor-level analysis across equity, hybrid, and debt funds.
- Time horizon: 5+ years of daily/nav and monthly AUM data where available.
- Deliverables: Analytical models, dashboard prototypes, and executive recommendations.

Problem Statement

Asset managers and product owners lack a consolidated, metrics-driven view of mutual fund performance and investor behavior. Disparate data sources and inconsistent calculations produce friction in performance attribution and risk monitoring. The challenge: develop a repeatable analytics framework that delivers reliable performance metrics, risk indicators, and investor cohort insights to inform product strategy and portfolio decisions.

- Inconsistent NAV histories and missing data points across vendors.
- Limited visibility into investor-level persistence (SIP continuity, redemptions).
- No standardized implementation of advanced risk metrics (Sortino, CVaR) for daily monitoring.

Project Objectives

Objective 1

Standardize performance calculation (CAGR, total return) across funds and vintages.

Objective 2

Implement risk analytics (Sharpe, Sortino, VaR, CVaR, max drawdown) with daily rolling windows.

Objective 3

Develop investor-level analytics: cohort segmentation, SIP continuity, churn analysis.

Objective 4

Deliver an interactive Power BI dashboard for monitoring and stakeholder reporting.

Success criteria: reproducible calculations, dashboard adoption by PMs, and insights driving at least two product or risk actions within 6 months.

Data Sources

The analysis synthesizes data from multiple verified sources to ensure completeness and accuracy:

- Vendor NAV histories (daily NAVs) — primary source for return calculations.
- Registrar & Transfer Agent (RTA) transactional records — investor-level SIP, redemption, and SIP stop/start events.
- Fund fact sheets and portfolio holdings snapshots — for attribution and concentration analysis.
- Market benchmarks and risk-free rates (overnight/1M T-bill) — for Sharpe/Sortino baselines.
- Macroeconomic series (inflation, GDP growth) — contextual overlays for scenario analysis.

Data ingestion used secure APIs and ETL pipelines into a centralized analytics store with schema versioning and provenance metadata.

Data Cleaning Process

High-quality analytics depend on rigorous preprocessing. Key cleaning steps:

1. Timestamp alignment: converted all dates to a single timezone and business-day calendar; forward-filled NAVs for market holidays when appropriate.
2. Missing values: imputed short gaps (<3 days) via linear interpolation; flagged and separately handled longer gaps.
3. Outlier detection: winsorized NAV returns beyond 5 standard deviations and validated against vendor reconciliations.
4. Normalization: converted price-adjusted returns to total return series accounting for dividends/distributions.
5. Investor records: deduplicated investor IDs, normalized SIP frequency (monthly/quarterly), and reconciled AMCs' investor buckets.

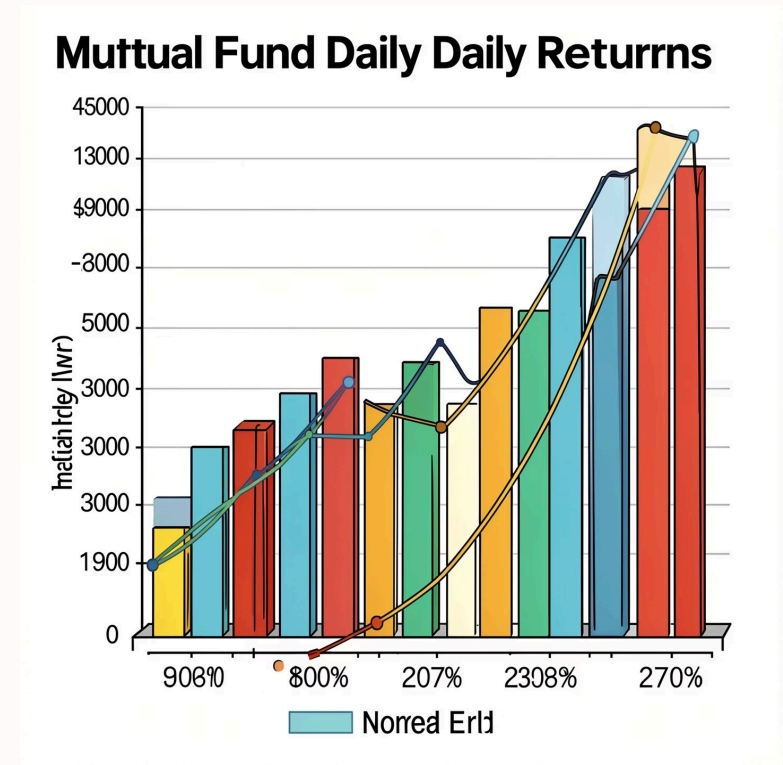
An audit trail preserves raw files and transformation logs. Data quality checks include daily reconciliation reports and exception dashboards surfaced in Power BI.

Exploratory Data Analysis

Initial EDA focused on distributional properties and relationships across funds. Primary observations:

- Return distributions vary materially by fund category: equity funds show heavier tails and higher volatility compared to debt funds.
- Significant autocorrelation in short-term returns for some hybrid funds, suggesting smoothing or illiquid asset effects.
- Cross-sectional correlations reveal clusters of funds with similar factor exposures (large-cap growth, value tilt).

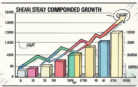
These insights guided model choices for risk-adjusted metrics and informed cohort definitions for investor analysis.



Histogram: daily return distribution (example fund)

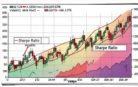
Performance Analytics

This section details methodology and representative outputs for the principal performance metrics used to evaluate funds.



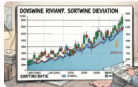
CAGR

Compound Annual Growth Rate computed from geometric returns over the measurement horizon. Used to compare long-term performance across funds with different vintages.



Sharpe Ratio

Return in excess of the risk-free rate divided by volatility (standard deviation of returns). Calculated on annualized, excess return basis. Rolling windows (1Y, 3Y) used for time-varying assessment.



Sortino Ratio

Similar to Sharpe but uses downside deviation, focusing on harmful volatility. Preferred for strategies with asymmetrical risk profiles.



Maximum Drawdown

Measured peak-to-trough decline over the period. Used for stress-testing and capital at risk considerations. Complemented by duration of drawdown events.

Example table and charts (placeholders) were created in Power BI for side-by-side comparison with peer benchmarks.

Risk Analytics

Advanced risk metrics provide deeper insights into tail exposures and performance persistence.

Value at Risk (VaR)

Parametric and historical VaR computed at 95% and 99% confidence levels on daily returns. Backtesting performed using Kupiec tests and conditional coverage.

Conditional VaR (CVaR)

Expected shortfall beyond the VaR threshold, offering a measure of average loss in worst-case tails. Important for stress planning and capital buffers.

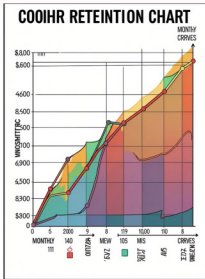
Rolling Sharpe Ratio

Time-series of Sharpe Ratio computed on rolling windows (90d, 252d, 3Y). Highlights regime shifts and persistence of risk-adjusted performance.

Charts include VaR backtesting plots, CVaR sensitivity to confidence level, and rolling Sharpe time-series for peer comparison.

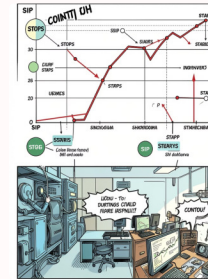
Investor Analytics

Investor behavior analysis translates fund performance into real-world product outcomes:



Cohort Analysis

Grouped investors by month/year of first investment to measure retention, average AUM progression, and churn. Key metrics: 3-, 6-, 12-month retention and average AUM per cohort.



SIP Continuity Analysis

Analyzed SIP on/off behavior, gaps in contributions, and factors correlated with SIP stoppage (drawdown magnitude, negative returns, life events). Identified segments with high persistence and at-risk cohorts.

Outputs include retention heatmaps, cohort lifetime value estimates, and predictive signals for SIP discontinuation used for targeted retention campaigns.